Industrial Internship Report on

”Crop and Weed Detection”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

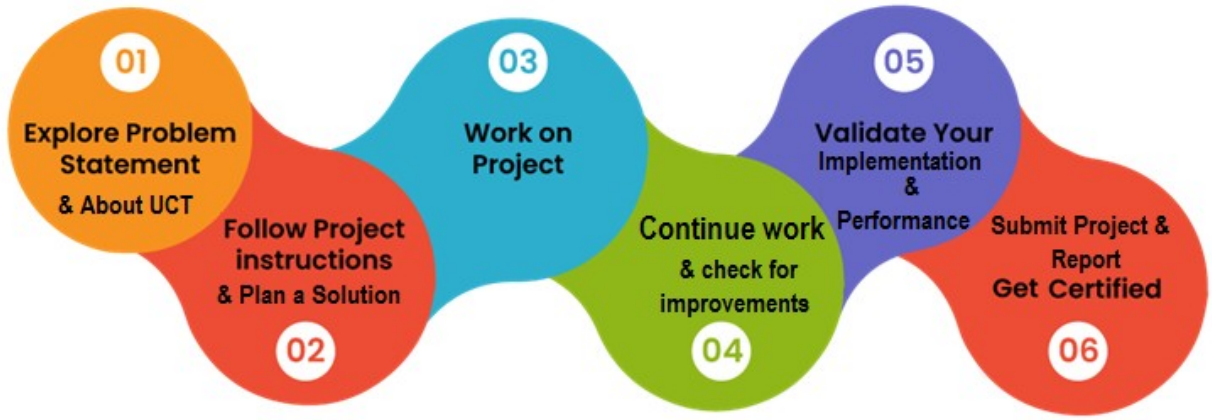
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

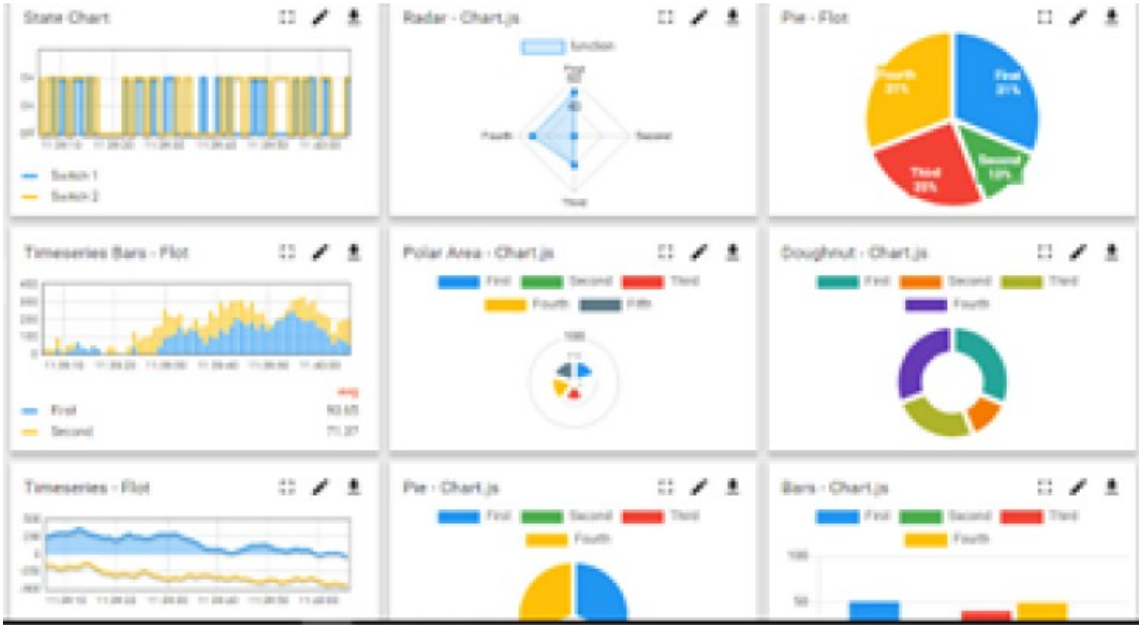
i. UCT IoT Platform (uct Insight)

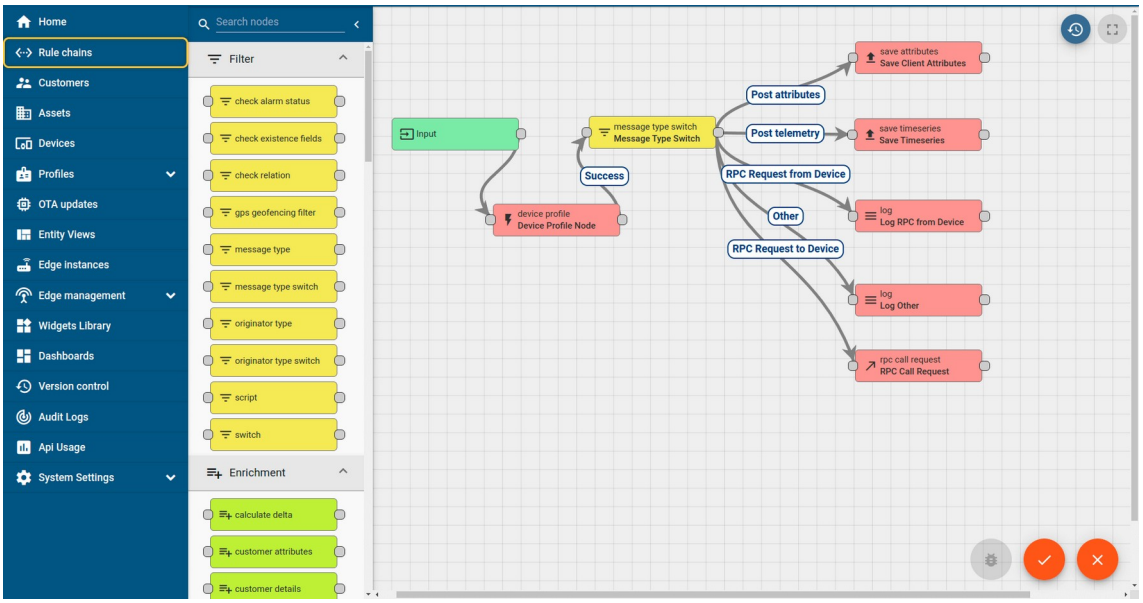
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

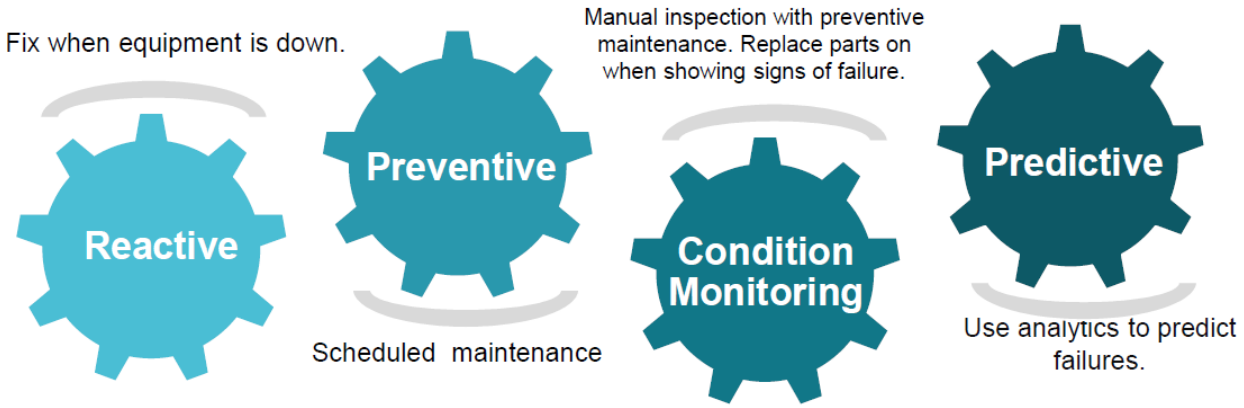
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

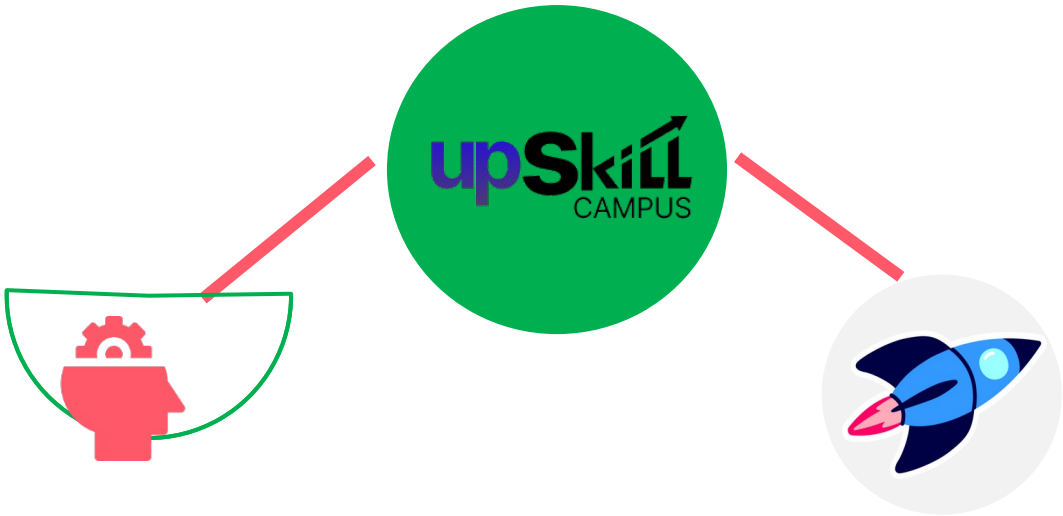
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

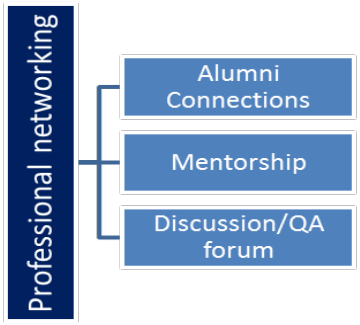
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

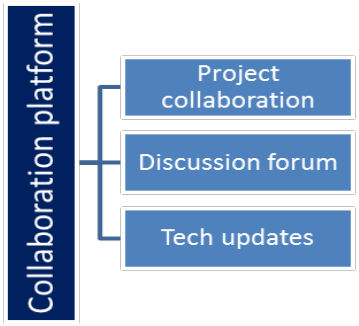


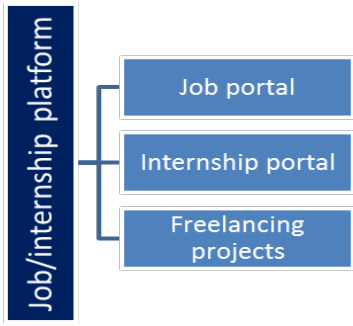
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Ultralytics YOLO Documentation – https://docs.ultralytics.com

[2] GeeksforGeeks – Machine Learning & Object Detection Articles

[3] OpenCV Documentation – https://opencv.org

2.6 Glossary

Terms	Acronym
YOLO	You Only Look Once – Object Detection Algorithm
Bounding Box	Rectangle around detected object
Precision	Accuracy of correct positive predictions
Recall	Ability to detect all relevant objects
Dataset	Collection of images used for training

3 Problem Statement

In the assigned problem statement

Weed is an unwanted thing in agriculture. Weed use the nutrients, water, land and many more things that might have gone to crops. Which results in less production of the required crop. The farmer often uses pesticides to remove weed which is also effective but some pesticides may stick with crop and may causes problems for humans.

4 Existing and Proposed solution

Existing Solution

- Manual weed removal (time-consuming)
- Uniform pesticide spraying (inefficient)
- Basic machine learning models (low accuracy)

Limitations

- High labour cost
- Excess chemical usage
- No automation
- Low precision

Proposed Solution

We developed a YOLO-based object detection model that**:**

- Detects crops and weeds from images
- Draws bounding boxes around weeds
- Can be integrated with smart spraying systems

4.1 Code submission (Github link)

Code:
https://github.com/bhupathi3812-m/upskillcampus/blob/main/src/CropAndWeedDetection.py

4.2 Report submission (Github link) :

4.3 Report link:

5 Proposed Design/ Model

Workflow

1. Data Collection (1300 images)
2. Annotation (YOLO format)
3. Data Cleaning & Validation
4. Train-Test Split
5. Model Training (YOLOv8)
6. Evaluation (Precision, Recall, mAP)
7. Prediction & Visualization

5.1 High Level Diagram

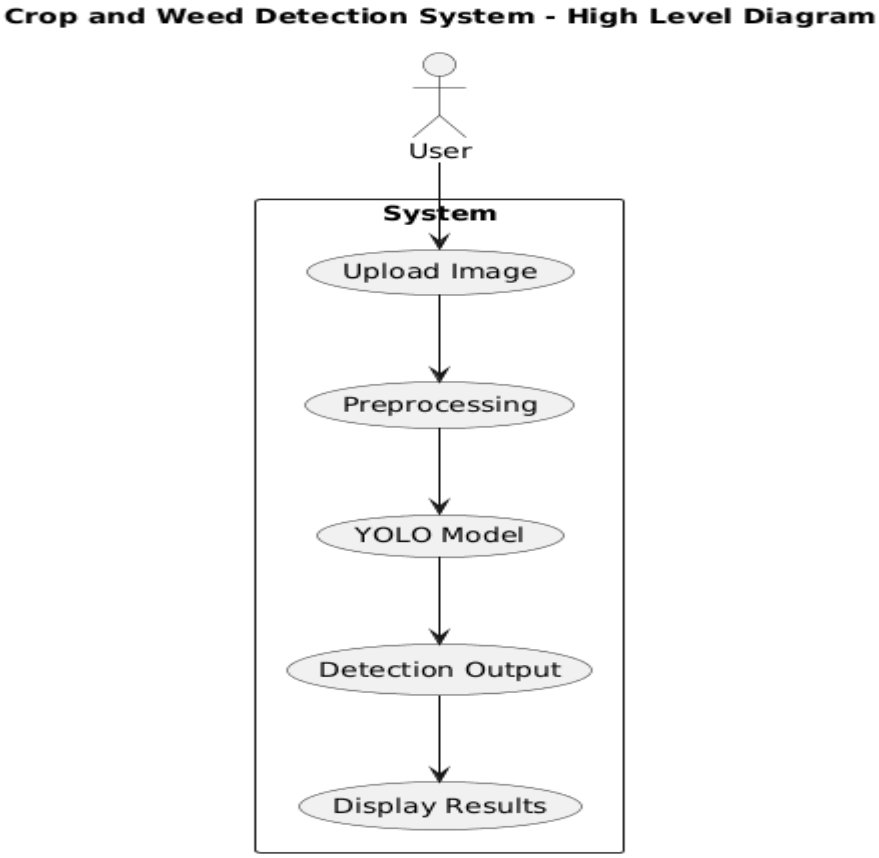


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

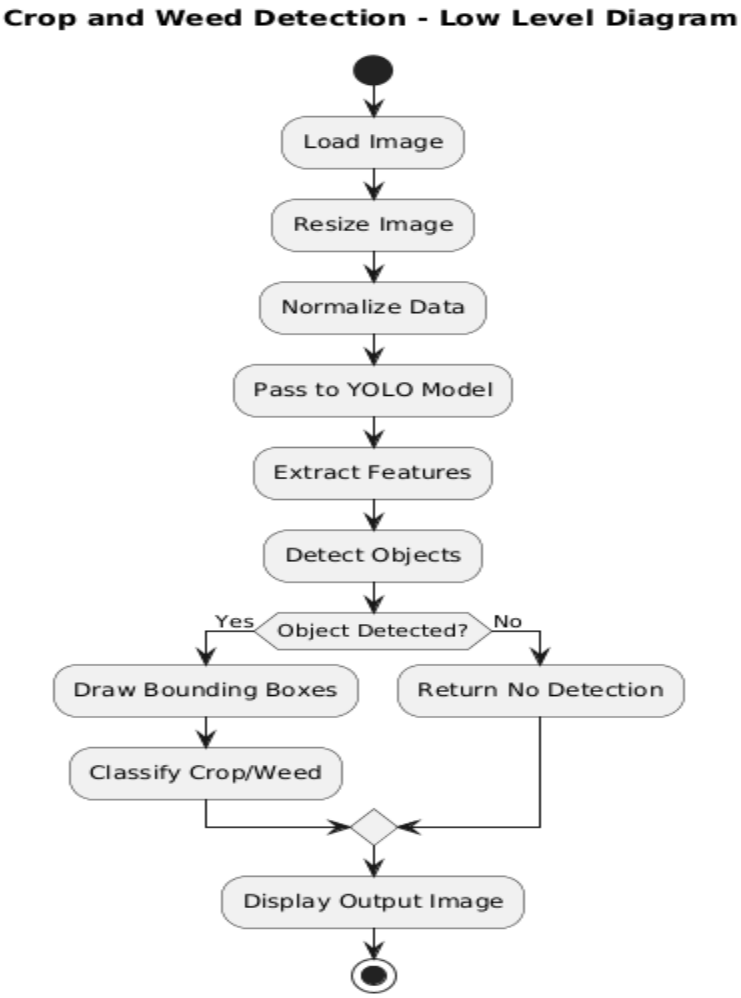


Figure 2:Low level diagram

5.3 Interfaces

Data flow.

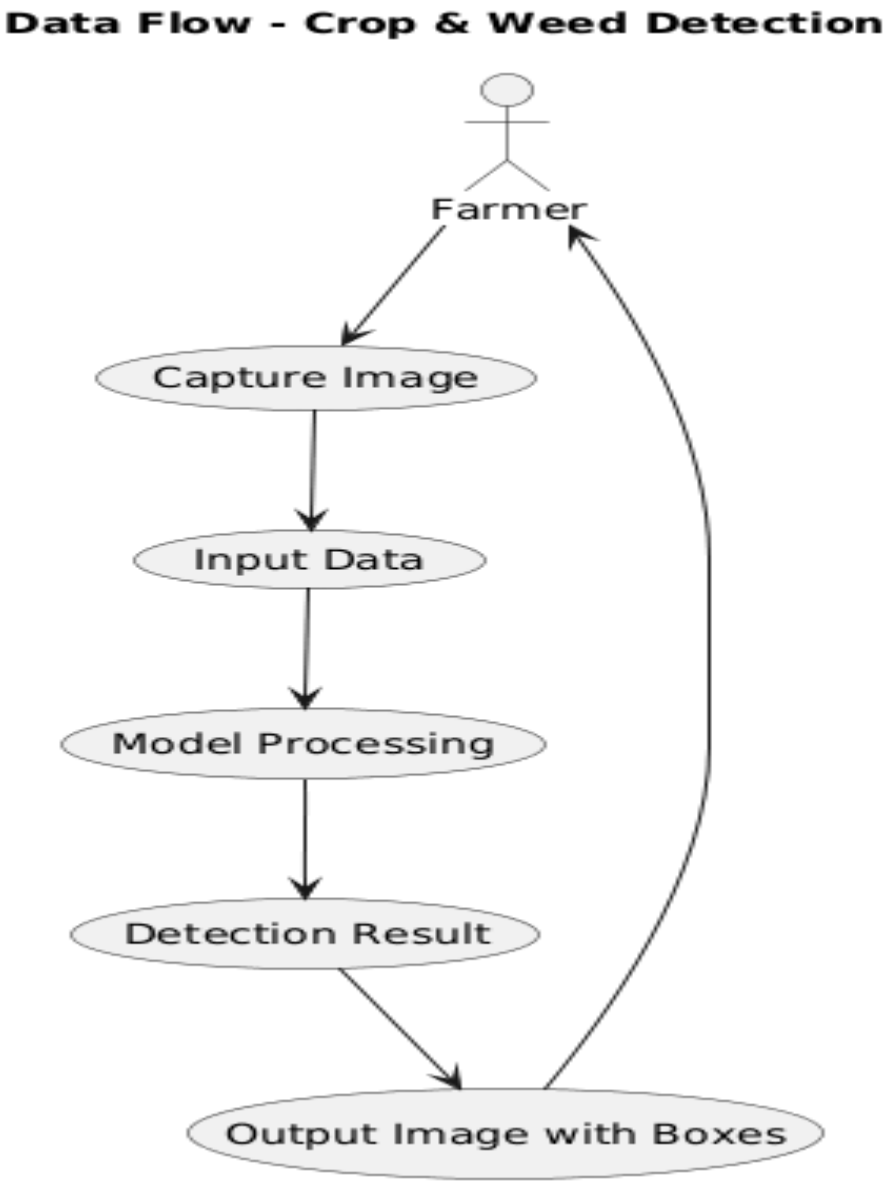


Figure 3:Dataflow

6 Performance Test

The performance testing phase evaluates the effectiveness and reliability of the proposed crop and weed detection system in real-world scenarios. This ensures that the system is not only theoretically correct but also practically usable in agricultural applications.

Identified Constraints

The following constraints were considered:

- **Accuracy**: Correct classification of crop vs weed
- **Speed (Inference Time)**: Time taken to process each image
- **Hardware Limitation**: Training and testing performed on CPU (limited processing power)
- **Dataset Size**: Limited dataset (1300 images)
- **Memory Usage**: Model should be lightweight for deployment
- **Real-world Variability**: Lighting, background, and plant variations

Handling of Constraints

- Used **YOLOv8 model** for efficient and fast object detection
- Applied **data cleaning and validation** to improve data quality
- Optimized image size (**512x512**) to balance speed and accuracy
- Used **batch processing** during training
- Evaluated model using standard metrics (precision, recall, mAP)

6.1 Test Plan/ Test Cases

Test Case	Description	Expected Result
TC1	Detect crop in image	Crop correctly identified
TC2	Detect weed in image	Weed correctly identified
TC3	Mixed image (crop + weed)	Both detected with bounding boxes
TC4	Empty/invalid image	No detection
TC5	Different lighting conditions	Stable detection performance

6.2 Test Procedure

Load trained YOLO model (`best.pt`)

- Provide input test images from validation dataset
- Run prediction using model
- Observe output images with bounding boxes
- Compare predictions with actual labels
- Calculate evaluation metrics (precision, recall, mAP)

6.3 Performance Outcome

The model achieved the following results:

- **Precision**: 82%
- **Recall**: 79%
- **mAP@50**: 85%
- **mAP@50-95**: 58%
- **Inference Time**: ~90 ms per image

7 My learnings

During this internship, I learned:

- Object Detection using YOLO
- Data preprocessing and annotation
- Model training and evaluation
- Handling real-world dataset issues
- Debugging environment and dependency errors
- Structuring ML projects professionally

This internship improved my:

- Problem-solving skills
- Practical ML knowledge
- Confidence in building real-world applications

8.Future work scope

Integrate with IoT-based smart sprayer system

- Deploy as **mobile or web application**
- Improve accuracy with larger dataset
- Use real-time video detection
- Apply drone-based monitoring